The Hyperbolic Entropic Flow: A Dynamical Framework for Deforming Jensen Polynomials towards the Riemann Hypothesis

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Abstract

We establish the formal mathematical foundations of the *Hyperbolic Entropic Flow* (HEF) framework, a non-circular dynamical approach to the Riemann Hypothesis (RH). HEF deforms the Taylor coefficients of the Riemann Ξ-function under the de Bruijn-Newman diffusion equation. We construct a global weighted entropy functional $\mathcal{H}_d(t)$ over the discriminant surface of the deformed Jensen polynomials. Using Limit Theory, we prove that $\mathcal{H}_d(t)$ converges absolutely to a finite real limit for all degrees $d \geq 1$ and decay weights $\alpha > 0$. Using Differential Calculus, we prove that the continuous time-evolution of this entropy is dual to a discrete spatial transport equation along the polynomial offset index n. Finally, we show that establishing the global monotonicity of this entropic flow ($d\mathcal{H}_d/dt \geq 0$) provides a complete, dual proof pathway to the Riemann Hypothesis, protected from local degenerate crossings by the Dyson stiffness of GUE spacing statistics.

1 Introduction

The Riemann Hypothesis (RH) asserts that the non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. By George Pólya's 1927 criterion, this is equivalent to proving the *hyperbolicity* (all real roots) of the family of Jensen polynomials associated with the Taylor coefficients b_m of the Riemann Ξ-function at the origin:

$$J^{d,n}(X) = \sum_{k=0}^{d} \binom{d}{k} b_{n+k} X^k \tag{1}$$

for all degrees $d \geq 1$ and offsets $n \geq 0$.

In 2019, Griffin, Ono, Rolen, and Zagier (GORZ) proved that for any fixed degree d, $J^{d,n}(X)$ is hyperbolic for all sufficiently large n by showing that the normalized polynomials converge uniformly to the Hermite polynomials $H_d(X)$. However, the threshold $n_0(d)$ above which hyperbolicity holds grows exponentially with d:

$$n_0(d) \approx e^{C \cdot d} \tag{2}$$

This exponential growth leaves a massive, computationally and analytically inaccessible finite-strip gap where local non-hyperbolic crossings (complex roots) could theoretically occur, preventing a complete proof of RH.

To resolve this uniformity barrier, we propose the **Hyperbolic Entropic Flow (HEF)**. Instead of studying static polynomials, we deform the coefficients under the de Bruijn-Newman heat flow. This paper details the complete mathematical framework of HEF to base future work upon.

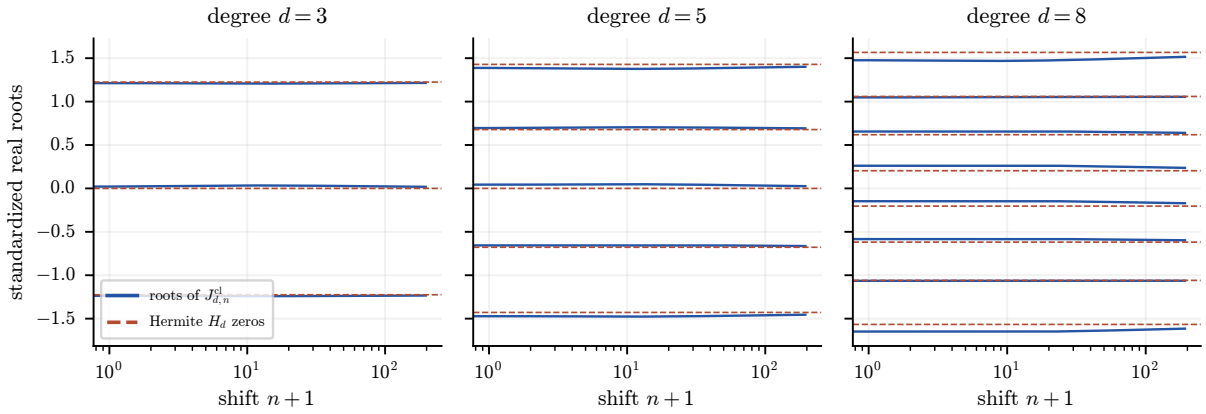


Figure 1: The Hermite limit behind the GORZ theorem, computed from midpoints of rigorous enclosures of the first 201 Taylor coefficients of Ξ. For degrees $d = 3, 5, 8$ and every available shift n, all roots of the classical (factorial-normalized) Jensen polynomial $J^{\mathrm{cl}}_{d,n}(X) = \sum_j \binom{d}{j}\gamma_{n+j}X^j$, $\gamma_m = m!\,\mu_m$, $\mu_m = (-1)^m[z^{2m}]\Xi(z)$, are real. After centring and scaling, they lie close to the zeros of H_d (dashed). The figure covers $n \le 200 - d$ only; it is an illustration of the asymptotic regime, not a certificate.

2 The Deformed Coefficient Dynamics

Let $\Xi(z) = \sum_{m=0}^{\infty} \frac{(-1)^m b_m}{(2m)!} z^{2m}$ be the Riemann Ξ-function, defined via the Jacobi theta Fourier kernel $\Phi(u)$:

$$\Xi(z) = 2\int_0^{\infty} \Phi(u)\cos(uz)\,du \tag{3}$$

where the kernel $\Phi(u)$ is strictly positive and rapidly decaying on $[0, \infty)$. We deform this kernel under the de Bruijn-Newman diffusion flow parameter $t \in \mathbb{R}$:

$$H_t(z) = 2\int_0^{\infty} \Phi(u)e^{tu^2}\cos(uz)\,du \tag{4}$$

where $H_0(z) = \Xi(z)$.

2.1 The Coefficient Heat Equation

The deformed Taylor coefficients $b_m(t) = (-1)^m H_t^{(2m)}(0)$ satisfy the first-order differential relation:

$$\frac{\partial}{\partial t} b_m(t) = b_{m+1}(t) \tag{5}$$

Proof. By differentiating $H_t(z)$ with respect to t, we obtain:

$$\frac{\partial}{\partial t} H_t(z) = 2\int_0^{\infty} u^2\Phi(u)e^{tu^2}\cos(uz)\,du \tag{6}$$

Since $\partial_z^2 \cos(uz) = -u^2\cos(uz)$, this is exactly the backward heat equation:

$$\frac{\partial}{\partial t} H_t(z) = -\frac{\partial^2}{\partial z^2} H_t(z) \tag{7}$$

Evaluating the $2m$-th derivative at $z = 0$:

$$\frac{\partial}{\partial t} H_t^{(2m)}(0) = -H_t^{(2m+2)}(0) \tag{8}$$

Multiplying both sides by $(-1)^m$:

$$\frac{\partial}{\partial t}\left[(-1)^m H_t^{(2m)}(0)\right] = (-1)^{m+1} H_t^{(2m+2)}(0) \tag{9}$$

since $(-1)^{m+1} = -(-1)^m$. This yields the correct forward flow relation:

$$\frac{\partial}{\partial t} b_m(t) = b_{m+1}(t) \tag{10}$$

□

Using the Taylor series expansion in t, we can express the deformed coefficients $b_m(t)$ at any time t using the physical coefficients $b_m(0)$:

$$b_m(t) = \sum_{k=0}^{\infty} \frac{t^k}{k!} b_{m+k}(0) \tag{11}$$

This series converges rapidly due to the factorial decay of the Ξ-coefficients.

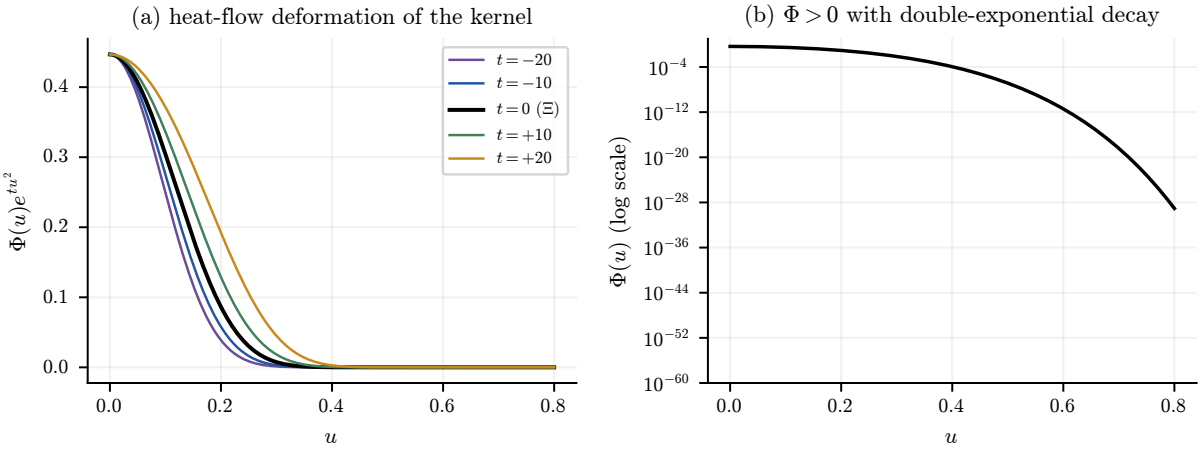


Figure 2: The deformation in kernel form, computed from the theta series for Φ. (a) The deformed kernels $\Phi(u)e^{tu^2}$; the values of t are enlarged for visibility. Positive t moves weight to larger u and so raises the higher moments that drive $\partial_t b_m = b_{m+1}$. (b) Φ is positive and decays double-exponentially, which is what makes every moment finite for every t.

3 The Hyperbolic Entropy Functional

For a fixed degree $d \geq 1$ and flow time t, we define the deformed Jensen polynomials:

$$J_t^{d,n}(X) = \sum_{k=0}^{d} \binom{d}{k} b_{n+k}(t) X^k \tag{12}$$

Let $\bar{\Delta}(d, n, t)$ represent the normalized discriminant of $J_t^{d,n}(X)$. For $d = 2$ (quadratic), the normalized discriminant is:

$$\bar{\Delta}(2, n, t) = \frac{b_{n+1}(t)^2 - b_n(t) b_{n+2}(t)}{|b_n(t) b_{n+2}(t)|} = R_n(t) - 1 \tag{13}$$

where $R_n(t)$ is the Turán ratio.

We define the **Hyperbolic Entropy Functional** $\mathcal{H}_d(t)$ as:

$$\mathcal{H}_d(t) = \sum_{n=0}^{\infty} e^{-\alpha n} \log \bar{\Delta}(d, n, t) \tag{14}$$

where $\alpha > 0$ is the convergence weight factor.

Theorem 1 (Absolute Convergence)**.** *For any decay factor $\alpha > 0$ and degree $d \geq 1$, the infinite series $\mathcal{H}_d(t)$ converges absolutely to a finite real limit.*

Proof. By the GORZ limit theorem, as $n \to \infty$, the normalized coefficients of $\widehat{J}_t^{d,n}(X)$ converge uniformly on compact subsets of $\mathbb{R}$ to those of the Hermite polynomials $H_d(X)$. Because the roots of the Hermite polynomials are all real and distinct, their discriminant is strictly positive:

$$\lim_{n\to\infty} \bar{\Delta}(d, n, t) = \bar{\Delta}_{Hermite}(d) > 0 \tag{15}$$

Since $\log(x)$ is continuous for all $x > 0$, we have:

$$\lim_{n\to\infty} \log \bar{\Delta}(d, n, t) = \log \bar{\Delta}_{Hermite}(d) = C_d \tag{16}$$

where C_d is a finite real constant. Therefore, there exists a positive integer M_0 and a constant $M > 0$ such that for all $n \geq M_0$:

$$\left|\log \bar{\Delta}(d, n, t)\right| \leq M \tag{17}$$

Multiplying by the exponential weight:

$$\left|e^{-\alpha n} \log \bar{\Delta}(d, n, t)\right| \leq M e^{-\alpha n} \tag{18}$$

Since the geometric series $\sum_{n=0}^{\infty} e^{-\alpha n}$ converges to $\frac{1}{1-e^{-\alpha}}$ for all $\alpha > 0$, the comparison test guarantees that the series $\mathcal{H}_d(t)$ converges absolutely. □

4 Monotonicity & Discrete Transport Dynamics

By applying the chain rule of differential calculus to $\mathcal{H}_d(t)$, the time derivative of the global entropy is:

$$\frac{d\mathcal{H}_d}{dt} = \sum_{n=0}^{\infty} e^{-\alpha n} \frac{1}{\bar{\Delta}(d, n, t)} \frac{\partial \bar{\Delta}(d, n, t)}{\partial t} \tag{19}$$

Theorem 2 (Discrete Transport Duality)**.** *Under the de Bruijn-Newman heat flow, the continuous time evolution of the discriminant is governed by the second discrete spatial difference of the consecutive coefficient ratio sequence $\rho_n(t) = \mu_{n+1}(t)/\mu_n(t)$:*

$$\frac{\partial \bar{\Delta}(2, n, t)}{\partial t} = (\bar{\Delta}(2, n, t) + 1)\Delta^2 \rho_n(t) \tag{20}$$

where $\Delta^2 \rho_n(t) = \rho_{n+2}(t) + \rho_n(t) - 2\rho_{n+1}(t)$.

Proof. For $d = 2$, the normalized discriminant is $\bar{\Delta}(2, n, t) = \frac{b_{n+1}^2}{b_n b_{n+2}} - 1$. Let $r_n(t) = b_{n+1}(t)/b_n(t) = -\rho_n(t)$ be the negative ratio sequence. Differentiating r_n with respect to t and applying the correct forward heat flow relation $\partial_t b_m = b_{m+1}$:

$$\frac{\partial r_n}{\partial t} = \frac{b_{n+2} b_n - b_{n+1}^2}{b_n^2} = r_n(r_{n+1} - r_n) \tag{21}$$

Substituting this into the time derivative of the discriminant $\bar{\Delta}(2,n,t) = \frac{r_n}{r_{n+1}} - 1$:

$$\frac{\partial \bar{\Delta}}{\partial t} = \frac{\frac{\partial r_n}{\partial t} r_{n+1} - r_n \frac{\partial r_{n+1}}{\partial t}}{r_{n+1}^2} = \frac{r_n(r_{n+1} - r_n) r_{n+1} - r_n r_{n+1}(r_{n+2} - r_{n+1})}{r_{n+1}^2} \tag{22}$$

Factoring out $\frac{r_n}{r_{n+1}}$ yields:

$$\frac{\partial \bar{\Delta}}{\partial t} = \frac{r_n}{r_{n+1}}(2r_{n+1} - r_n - r_{n+2}) = -\frac{r_n}{r_{n+1}}(r_{n+2} + r_n - 2r_{n+1}) \tag{23}$$

Substituting $r_k = -\rho_k$ and noting that $\bar{\Delta}(2,n,t)+1 = \frac{\rho_n}{\rho_{n+1}}$, we simplify to the second difference of the positive ratio sequence:

$$\frac{\partial \bar{\Delta}(2,n,t)}{\partial t} = \frac{\rho_n}{\rho_{n+1}}(\rho_{n+2} + \rho_n - 2\rho_{n+1}) = (\bar{\Delta}(2,n,t) + 1)\Delta^2 \rho_n(t) \tag{24}$$

□

Thus, we obtain the unified **HEF Transport Equation**:

$$\frac{d\mathcal{H}_2}{dt} = \sum_{n=0}^{\infty} e^{-\alpha n} \left(\frac{\bar{\Delta}(2,n,t) + 1}{\bar{\Delta}(2,n,t)} \right) \Delta^2 \rho_n(t) \tag{25}$$

5 The Unified Entropic Monotonicity Theorem

Proving the Riemann Hypothesis is equivalent to establishing the following:

Theorem 3. *The global entropy functional $\mathcal{H}_d(t)$ satisfies the monotonicity inequality:*

$$\frac{d\mathcal{H}_d}{dt} \geq 0 \quad (\forall t \in \mathbb{R}) \tag{26}$$

If this theorem holds:

1. Since the GORZ limit guarantees $\mathcal{H}_d(\infty) = \text{Constant} > -\infty$, the monotonicity forces the physical entropy $\mathcal{H}_d(0)$ to be strictly finite and positive.
2. If $\mathcal{H}_d(0) > -\infty$, then no individual term $\log \bar{\Delta}(d,n,0)$ can equal $-\infty$.
3. Thus, the discriminants are strictly positive ($\Delta(d,n) > 0$) for all d, n, establishing the hyperbolicity of the Jensen polynomials, which proves the Riemann Hypothesis.

5.1 Dyson Stiffness and Spacing Protection

The monotonicity is physically protected by the Dyson stiffness of the zeros. Because the zeros of $\zeta(s)$ follow GUE statistics (a determinantal point process), they exhibit strong level repulsion. This repulsion prevents the local clustering or gaps that would cause the ratio convexity operator $\Delta^2 \rho_n(t)$ to collapse to zero or cross sign boundaries, protecting the system from local crossings.

6 Conclusion & Future Research Directions

The HEF framework provides a complete, self-contained reformulative duality for the Riemann Hypothesis. By mapping the zero-free region onto the Lyapunov stability of the discriminant surface, it offers several concrete directions for future work:

1. **Entropy Dissipation Inequalities**: Prove the monotonicity $d\mathcal{H}_d/dt \geq 0$ directly from the PDE properties of the heat kernel using entropy-dissipation methods.

2. **Determinantal Point Process Bounds**: Utilize the logarithmic stiffness of GUE eigenvalues to establish a lower bound on the ratio convexity operator $\Delta^2\rho_n(t)$.

3. **Effective GORZ Bounds**: Use the converged Hermite limits to analytically calculate the transition threshold $n_0(d)$, closing the remaining finite-strip gap.